The Moment Generating Function (MGF)

Def^n: Given a random variable X, the moment-generating function $M_X(t)$ of its probability distribution is the expected value of e^{tX}. Expressed mathematically,

$$M_X(t) = E(e^{tX})$$

$$= \begin{cases} \sum_i e^{tx_i} p(x_i) & \text{discrete } x, \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx & \text{continuous } x. \end{cases}$$

Expanding e^{tX} as a power series in t, we obtain

$$e^{tX} = 1 + tX + \frac{t^2X^2}{2!} + \cdots + \frac{t^rX^r}{r!} + \cdots$$

On taking expectations,

$$M_X(t) = E(e^{tX}) = 1 + E(X)\cdot t + E(X^2)\cdot\frac{t^2}{2} + \cdots + E(X^r)\cdot\frac{t^r}{r!} + \cdots$$

$$= 1 + \mu_1'\cdot t + \mu_2'\cdot\frac{t^2}{2} + \cdots + \mu_r'\cdot\frac{t^r}{r!} + \cdots$$

Therefore $\frac{d^r}{dt^r} M_X(t)\Big|_{t=0} = \mu_r'$.

$(a+b)^n = a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \cdots + b^n$

Ex. Suppose that X has a binomial distribution $B(n,p)$. The moment-generating function $M_X(t)$ is

$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0,1,2,\dots,n$

$$M_X(t) = \sum_{x=0}^{n} e^{tx} p(x)$$

$$= \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x (1-p)^{n-x}$$

or, $M_X(t) = [pe^t + (1-p)]^n$.

$P(X=x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(1-\frac{1}{2}\right)^{3-x}, \quad x = 0,1,2,3$

$= \binom{3}{x} \left(\frac{1}{2}\right)^3 = \frac{1}{8}\binom{3}{x}, \quad x = 0,1,2,3$

$P(X=0) = \frac{1}{8}$

$P(X=1) = \frac{3}{8}, \quad P(X=2) = \frac{3}{8}$

$P(X=3) = \frac{1}{8}$

Taking the derivatives, we obtain

$$M_X'(t) = npe^t [1 + p(e^t - 1)]^{n-1} \quad \text{and} \quad M_X''(t) = npe^t(1 - p + npe^t)[1 + p(e^t - 1)]^{n-2}.$$

Thus $\mu_1' = \mu = M_X'(t)\big|_{t=0} = np$ and $\mu_2' = M_X''(t)\big|_{t=0} = np(1-p+np)$.

$\therefore$ The variance $\sigma^2 = \mu_2' - \mu'^2 = np(1-p)$.

Ex: Derive the moment generating function of a Poisson distribution $P(\lambda)$ and hence find the mean and variance.

Ex: The concentration of reactant in a chemical process is a random variable having probability distribution

$$f(r) = 6r(1-r), \quad 0 \le r \le 1,$$
$$= 0, \quad \text{otherwise}.$$

The profit associated with the final product is $P = \$1.00 + \$3.00R$. Find the expected value of P. What is the probability distribution of P?